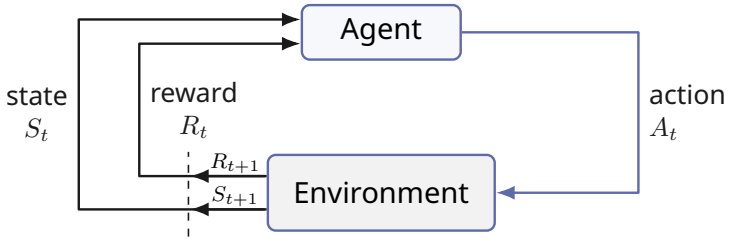
Advances in the Theory of

Offline Constrained Reinforcement Learning

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What Is Reinforcement Learning?



- **RL**: Maximize reward.
- **Constrained RL**: Maximize reward while satisfying safety constraints.

Problem Setup

CMDP protocol

The environment chooses an initial state $s_0 \in \mathcal{S}$.
For $t = 0, 1, 2, \ldots$:

1. Take action $a_t \sim \pi(\cdot \mid s_t) \in \Delta(\mathcal{A})$.
2. Receive reward $r(s_t, a_t)$.
3. Incur cost $c(s_t, a_t)$.
4. Advance to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$.

- Discounted returns:

$$J_r(\pi) = \mathbb{E}^\pi\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right], \qquad J_c(\pi) = \mathbb{E}^\pi\left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t)\right]$$

- Goal: $\max_\pi J_r(\pi)$ subject to $J_c(\pi) \leq \tau$.

Offline RL vs. Online RL

- **Online RL**: interact with the environment to collect trajectory data. (unsafe)
- **Offline RL**: use pre-collected trajectory data.

Today's Talk

Design computationally efficient (oracle efficient) and sample efficient algorithm for offline CMDP with large state space.

- **Function approximation**: e.g. $\pi \in \Pi$.
- **Plug-in idea:** estimate reward and cost from data, then solve

$$\max_{\pi \in \Pi} \ \widehat{J}_r(\pi) \qquad \text{subject to} \qquad \widehat{J}_c(\pi) \leq \tau.$$

Off-Policy Evaluation

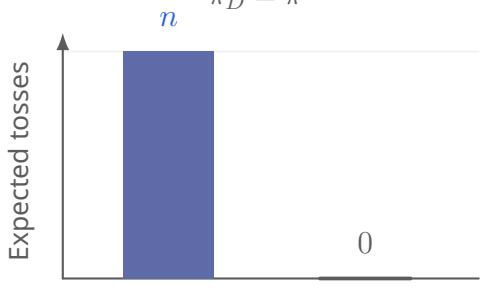
Given a policy π, how to estimate $J_r(\pi)$ and $J_c(\pi)$ from data?

Off-Policy Evaluation

Estimate p_A: $\widehat{p}_A = \dfrac{\#\{\text{heads from A}\}}{n\,\pi_D(A)}$

On-policy

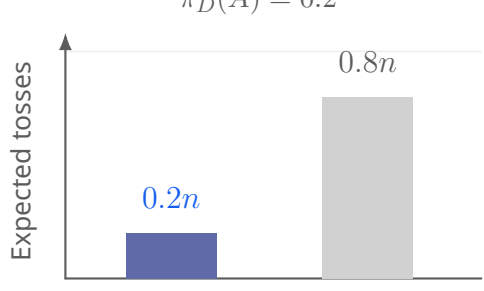
$\pi_D = \pi$



$|\widehat{p}_A - p_A| \lesssim \frac{1}{\sqrt{n}}$ (w.h.p.)

Off-policy

$\pi_D(A) = 0.2$



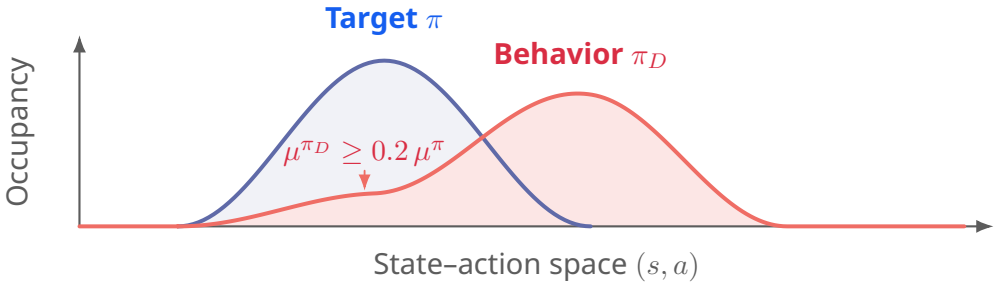
$|\widehat{p}_A - p_A| \lesssim \frac{5}{\sqrt{n}}$ (w.h.p.)

Occupancy Measures and Data Coverage

Definition (Occupancy Measures)

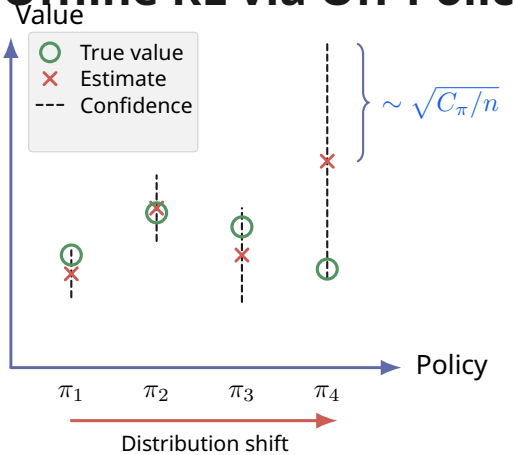
The occupancy measure of a policy π is defined as

$$\mu^{\pi}(s,a) := (1-\gamma)\sum_{t=0}^{\infty}\gamma^{t}\,P^{\pi}(s_t = s,\ a_t = a).$$



$$C_{\pi} := \max_{s,a}\frac{\mu^{\pi}(s,a)}{\mu^{\pi_D}(s,a)} = 5, \qquad \left|\widehat{J}(\pi) - J(\pi)\right| \lesssim \sqrt{\frac{C_{\pi}}{n}} \quad \text{(w.h.p.)}.$$

Offline RL via Off-Policy Evaluation



- **Best policy:** π_2
- **Best according to the estimate:**

$$\pi_4 = \operatorname*{argmax}_{\pi} \widehat{J}(\pi).$$

- **Guarantee**

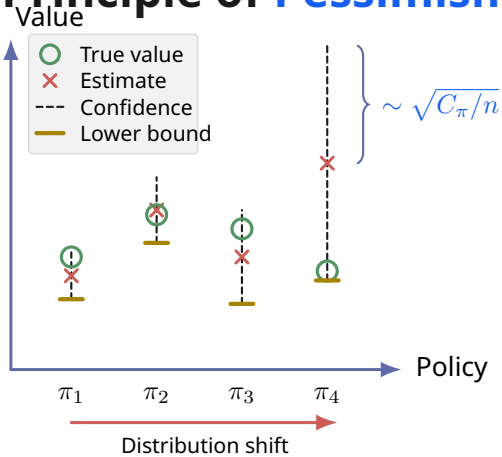
$$J_r(\widehat{\pi}) \geq J_r(\pi^*) - \max_{\pi} \sqrt{\frac{C_\pi}{n}}.$$

C_{π_4} is very large and can ruin the performance.

All-policy coverage

The dataset should cover all policies $\pi \in \Pi$ so that C_π is small across all π.

Principle of Pessimism



- **Best policy:** π_2
- **Best according to the lower bound:**

$$\pi_2 = \operatorname*{argmax}_\pi \widehat{J}(\pi) - \mathcal{O}(\sqrt{C_\pi/n}).$$

- **Guarantee**

$$J_r(\widehat{\pi}) \geq J_r(\pi^*) - \sqrt{\frac{C_{\pi^*}}{n}}.$$

Only requires data to cover the **best policy**!

One-policy coverage

Only requires data to cover the **best policy** so that C_{π^*} is small.

Information-Theoretic Algorithm

- For each π, off-policy evaluation:

$$J_r(\pi) \in \left[\widehat{L}_r(\pi), \widehat{U}_r(\pi)\right], \qquad J_c(\pi) \in \left[\widehat{L}_c(\pi), \widehat{U}_c(\pi)\right].$$

Offline RL

$$\max_{\pi \in \Pi} \quad \widehat{L}_r(\pi)$$

Offline constrained RL

$$\max_{\pi \in \Pi} \quad \widehat{L}_r(\pi)$$

$$\text{subject to} \quad \widehat{U}_c(\pi) \leq \tau$$

Question

How can we design a **computationally efficient** algorithm?

Actor–Critic for Offline RL

- $Q^\pi(s,a) = \mathbb{E}^\pi[\sum_{k=0}^{\infty} \gamma^k r(s_k,a_k) \mid s_0 = s,\ a_0 = a]$.

Exact policy iteration

- Actor: $\pi_{t+1}(s) \leftarrow \operatorname{argmax}_a Q_t(s,a)$.
- Critic: $Q_{t+1} \leftarrow Q^{\pi_{t+1}}$.

Actor–critic

- Actor: $\pi_{t+1} \leftarrow \mathrm{PO}(\pi_t, \widehat{Q}_t)$.
- Critic: $\widehat{Q}_{t+1} \leftarrow \mathrm{OPE}^{\mathrm{pess}}(\pi_{t+1}; r)$.

$$J_r(\pi^*) - J_r(\widehat{\pi}) \lesssim \sqrt{C_{\pi^*}}\, \varepsilon_{\mathrm{stat}} + \varepsilon_{\mathrm{opt}}.$$

Critic limitation under general function approximation

- **PSPI** (NeurIPS 2021), **ATAC** (ICML 2022): Bellman-error minimax critic.
- **A-Crab** (NeurIPS 2023): importance-weighted minimax critic.

Primal-Dual for Offline Constrained RL

- Lagrangian formulation:

$$\max_{\pi} \min_{\lambda \geq 0} \mathcal{L}(\pi, \lambda), \qquad \mathcal{L}(\pi, \lambda) = J_r(\pi) + \lambda\big(\tau - J_c(\pi)\big).$$

- Primal–dual algorithm:

$$\begin{aligned} \text{Actor:} \quad & \pi_{t+1} \leftarrow \mathrm{PO}(\pi_t, \widehat{Q}_t) \\ \text{Critic:} \quad & \widehat{Q}_{t+1} \leftarrow \mathsf{OPE}^{\mathsf{pess}}(\pi_{t+1}; r - \lambda_t c) \\ \text{Dual:} \quad & \lambda_{t+1} \leftarrow \big[\lambda_t + \eta\big(\widehat{J}_c(\pi_{t+1}) - \tau\big)\big]_+. \end{aligned}$$

- **Obstacle:** Sign errors in $\widehat{J}_c(\pi_t) - \tau$ can misdirect dual updates.

Hong et al. (2024)

Requires all-policy data coverage for estimating the dual update direction.

Constrained RL via Linear Programming

$$(1-\gamma)J_r(\pi) = \langle \mu^\pi, r \rangle, \qquad (1-\gamma)J_c(\pi) = \langle \mu^\pi, c \rangle.$$

Linear program

Let ρ_0 be the initial-state distribution.

$$\begin{aligned} \max_{\mu \geq 0} \quad & \langle \mu, r \rangle \\ \text{subject to} \quad & \langle \mu, c \rangle \leq (1-\gamma)\tau, \\ & \sum_a \mu(s,a) = (1-\gamma)\rho_0(s) + \gamma \sum_{x,b} P(s \mid x, b)\mu(x,b), \quad \forall s. \end{aligned}$$

Policy extraction: $\pi^\star(a \mid s) = \dfrac{\mu^\star(s,a)}{\sum_b \mu^\star(s,b)}$ (normalize within each state).

From the Lagrangian to Offline Data

- **Affine in the occupancy μ:**

$$\mathcal{L}(\mu; V, \lambda) = \underbrace{\mathbb{E}_{(s,a)\sim\mu}[g_{V,\lambda}(s,a)]}_{\text{linear in } \mu} + \underbrace{b_{V,\lambda}}_{\text{independent of } \mu}$$

- **Change of variables:** $\mu = w\mu_D$, $\mu_D = \mu^{\pi_D}$.

$$\mathcal{L}(w\mu_D; V, \lambda) = \mathbb{E}_{(s,a)\sim\mu_D}[w(s,a)\, g_{V,\lambda}(s,a)] + b_{V,\lambda}$$

- **Empirical average:** $\mathcal{D} = \{(s_i, a_i, r_i, c_i, s_i')\}_{i=1}^n$.

$$\widehat{\mathcal{L}}(w; V, \lambda) = \frac{1}{n}\sum_{i=1}^{n} w(s_i, a_i)\, \widehat{g}_{V,\lambda}^{(i)} + b_{V,\lambda}$$

$\widehat{g}_{V,\lambda}^{(i)}$: computed from observed transition i.

Linear Programming + Pessimism

- **Optimal coverage:** let C^* be an upper bound on the optimal weights.

$$w^* = \frac{\mu^{\pi^*}}{\mu_D}, \qquad \|w^*\|_\infty = C_{\pi^*} \leq C^*.$$

- **Cap importance weights:** optimize over bounded function classes.

$$\max_{\substack{w \in \mathcal{W} \\ 0 \leq w \leq C^*}} \min_{V \in \mathcal{V}} \min_{\lambda \in [0,B]} \widehat{\mathcal{L}}(w; V, \lambda).$$

- **Pessimism intuition:** the target policy π should not deviate too much from the behavior policy π_D.
- **Policy extraction:** direct normalization needs the unknown μ_D.

$$\widehat{\pi}(a \mid s) = \frac{\widehat{w}(s,a)\, \mu_D(s,a)}{\sum_b \widehat{w}(s,b)\, \mu_D(s,b)}.$$

Weighted Behavior Cloning

- **Exact weighted cloning:** recover an optimal policy $\pi^* \in \Pi$.

$$\pi^* \in \operatorname*{argmax}_{\pi \in \Pi} \mathbb{E}_{(s,a)\sim\mu_D}[w^*(s,a) \log \pi(a \mid s)] \,.$$

- **From data:** give actions with larger weights more influence.

$$\widehat{\pi} \in \operatorname*{argmax}_{\pi \in \Pi} \frac{1}{n} \sum_{i=1}^{n} \widehat{w}(s_i, a_i) \log \pi(a_i \mid s_i).$$

OptiDICE (Lee et al., 2021) COptiDICE (Lee et al., 2022)

Negative Result

Proposition (Ko & Hong 2026)

There exists a problem instance where:

- $\widehat{w}$ is optimal: policy $\widehat{\pi}$ extracted achieves $J_r(\widehat{\pi}) = J_r(\pi^*)$.
- Population weighted behavior cloning gives optimal policy:

$$\pi^* = \operatorname*{argmax}_{\pi \in \Pi} \mathbb{E}_{(s,a)\sim\mu_D}[\widehat{w}(s,a) \log \pi(a \mid s)].$$

- But empirical behavior cloning fails for all finite dataset of size n:

$$\widehat{\pi}_n = \operatorname*{argmax}_{\pi \in \Pi} \frac{1}{n} \sum_{i=1}^{n} \widehat{w}(s_i, a_i) \log \pi(a_i \mid s_i), \qquad P(J_r(\pi^*) - J_r(\widehat{\pi}_n) \geq 3/7) = 1/2.$$

Lagrangian Decomposition

- Easy to extract π from μ. Hard to extract π from w.
- However μ is converted to importance weight w for estimating Lagrangian.
- **Idea:** keep a copy of μ: $\nu = \mu$ (Lagrangian decomposition).

$$\begin{aligned} \max_{\mu \geq 0} \quad & \langle \mu, r \rangle \\ \text{subject to} \quad & \langle \mu, c \rangle \leq (1-\gamma)\tau, \\ & \textstyle\sum_a \mu(s,a) = (1-\gamma)\rho_0(s) + \gamma \sum_{x,b} P(s \mid x, b)\mu(x,b), \quad \forall s. \end{aligned}$$

$$\begin{aligned} \max_{\mu \geq 0,\ \nu \geq 0} \quad & \langle \mu, r \rangle \\ \text{subject to} \quad & \langle \mu, c \rangle \leq (1-\gamma)\tau, \\ & \textstyle\sum_a \nu(s,a) = (1-\gamma)\rho_0(s) + \gamma \sum_{x,b} P(s \mid x, b)\mu(x,b), \quad \forall s \\ & \mu = \nu \end{aligned}$$

Linear Programming + Policy Optimization

- Parameterize ν by a policy π using $\pi(a \mid s) = \nu(s,a) / \sum_b \nu(s,b)$.
- Reparameterize μ by $w = \mu/\mu_D$.

Reduced empirical Lagrangian

$$\widehat{\mathcal{L}}(w, \pi; Q, \lambda) = (1-\gamma)\big(\mathbb{E}_{s\sim\rho_0}[Q(s,\pi)] + \lambda\tau\big) + \frac{1}{n}\sum_{i=1}^{n} w(s_i,a_i)\Big[r_i - \lambda c_i + \gamma Q(s_i', \pi) - Q(s_i,a_i)\Big].$$

$$\max_{w\in\mathcal{W}} \max_{\pi\in\Pi} \min_{Q\in\mathcal{Q}} \min_{\lambda\in\Lambda} \widehat{\mathcal{L}}(w, \pi; Q, \lambda).$$

Primal–Dual with Critic

$$
\begin{aligned}
\text{Weights:} \quad & w_t \leftarrow \mathrm{OLO}_{\mathcal{W}}(g_1, \dots, g_{t-1}) \\
\text{Actor:} \quad & \pi_t \leftarrow \mathrm{PO}(Q_1, \dots, Q_{t-1}) \\
\text{Dual:} \quad & \lambda_t \in \operatorname*{argmin}_{\lambda \in \Lambda} \widehat{\mathcal{L}}(w_t, \pi_t; Q_{t-1}, \lambda) \\
\text{Critic:} \quad & Q_t \in \operatorname*{argmin}_{Q \in \mathcal{Q}} \widehat{\mathcal{L}}(w_t, \pi_t; Q, \lambda_t), \\
\text{where} \quad & g_t = \nabla_w \widehat{\mathcal{L}}(\cdot, \pi_t; Q_t, \lambda_t).
\end{aligned}
$$

- **OLO**: online linear optimization. **PO**: policy optimization.

Guarantee

- **Informal guarantee:** the output policy $\widehat{\pi}$ satisfies

$$(1-\gamma)J_r(\widehat{\pi}) \geq (1-\gamma)J_r(\pi^*) - \epsilon_n,$$

$$(1-\gamma)J_c(\widehat{\pi}) \leq (1-\gamma)\tau + \epsilon_n.$$

where

$$\epsilon_n = O\left(C_{\pi^*}\sqrt{\frac{\log\big(|\mathcal{W}|\,|\Pi|\,|\mathcal{Q}|/\delta\big)}{n}}\right).$$

Recap

Algorithm classes for offline CMDP:

- **Actor–critic:** currently requires all-policy data coverage due to dual update.
  - Open problem: can we design an algorithm that works with single-policy data coverage?
- **Linear programming:**
  - Open problem: design computationally efficient algorithm for policy extraction from w?